

Replacement of a gas boiler in a 300–600 m² small commercial building (office, retail, light industrial) with a 30 kW peak heat demand, connecting to a new or existing low-temperature hydronic distribution system (35–45°C flow), serving 5–20 terminal units (fan coils or underfloor heating), with annual heating energy demand of 60,000–120,000 kWh/year, reducing carbon emissions by approximately 60–70% vs. gas at UK grid carbon intensity of 0.207 kgCO₂e/kWh (DESNZ 2023 Conversion Factors).

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£4,500.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

TOTAL ADDRESSABLE MARKET

Not computed

[D] Modelled

[B] Research

SUBSYSTEM MODULES

9

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source
- [D] Algorithmic estimate or LLM extraction
- [E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	PASS	Standards found	10 standards
3. Cost Ceiling Specified	PASS	Target provided	£4,500.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	9 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements ^[A]

1.1 Mission & Use Case

Mission

To deliver a compact, MCS-certified 30 kW R290 air-source heat pump for small UK commercial buildings, achieving COP e 3.5 at A2/W35 with a hydronic split architecture at a unit cost accessible to the SME building retrofit market. The product enables commercial building owners to decarbonise heating whilst qualifying for UK government financial incentive schemes.

Use Case

Replacement of a gas boiler in a 300–600 m² small commercial building (office, retail, light industrial) with a 30 kW peak heat demand, connecting to a new or existing low-temperature hydronic distribution system (35–45°C flow), serving 5–20 terminal units (fan coils or underfloor heating), with annual heating energy demand of 60,000–120,000 kWh/year, reducing carbon emissions by approximately 60–70% vs. gas at UK grid carbon intensity of 0.207 kgCO_e/kWh (DESNZ 2023 Conversion Factors).

1.2 Market Context & Timing ^[B]

The UK F-Gas Regulation (retained EU 517/2014, now SI 2022/Future SI) mandates phase-down of high-GWP refrigerants; R410A (GWP 2,088) equipment faces ban on servicing >40 tonne CO_e systems by 2030, creating forced replacement market. The Boiler Upgrade Scheme (BUS) extended to commercial in 2024 with £5,000–£6,000 grants per installation. Clean Heat Market Mechanism from 2025 places obligations on boiler manufacturers, stimulating demand pull. BEIS Heat Pump Ready programme invested £60m in 2022–2024 in supply chain development. IEA (2023 Heat Pumps report) identifies commercial <50 kW as fastest-growing segment globally at 18% CAGR. No MCS-certified R290 unit in the 25–35 kW class currently exists in the UK market as of Q1 2024.

1. M&E contractors installing commercial building retrofits (500–2,000 installs/year firms, MCS-registered). 2. Commercial property owners and facilities managers of buildings 200–800 m² (offices, GP surgeries, small hotels, schools). 3. Energy Service Companies (ESCOs) and heat-as-a-service providers deploying under BUS/CHMM incentive schemes. 4. Local authorities procuring decarbonisation of small civic buildings under Public Sector

Decarbonisation Scheme (PSDS). 5. M&E consulting engineers specifying plant for planning-compliant Part L 2021 new-build or retrofit projects.

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Daikin Europe	
Product	Daikin Altherma 3 H HT ECH,O – 16 kW + 16 kW cascade (32 kW total)
Pricing	£6,800–£8,200 supply-only (trade price, UK distributor 2023)
Tech Specs	R32 refrigerant (GWP 675), COP 3.51 at A2/W35 (EN 14825), SCOP 3.83 (average climate, 55°C), flow temperature up to 70°C, 3-phase 400V, 9.5A max, ODU dimensions 1,345×930×390 mm, ODU weight 138 kg, noise 57 dB(A) at 3m, MCS certified, integrated electric heater 6 kW backup
STRENGTHS	WEAKNESSES
Established MCS certification, wide installer network, proven 70°C flow temperature for high-temperature retrofit, global spare parts availability, Daikin brand recognition with specifiers	R32 refrigerant faces future F-Gas restrictions; cascade architecture adds cost and complexity vs single-circuit; 32 kW achieved via two-unit cascade not single unit; no R290 option in commercial range; high price point exceeds BUS grant benefit
DIFFERENTIATION ANGLE	
Our R290 single-circuit 30 kW unit offers 60–70% lower GWP than R32, simpler installation (single refrigerant circuit), and £2,300–£3,700 lower supply price while matching COP performance	

Mitsubishi Electric

Product	Ecodan CAHV-R450YA-HPB Power Inverter (45 kW) – smallest commercial model in range
Pricing	£11,500–£14,000 supply-only (trade, UK 2023); no 30 kW R410A equivalent in range
Tech Specs	R410A refrigerant (GWP 2,088), rated heating capacity 45 kW at A7/W45, COP 4.22 at A7/W45 (EN 14825), SCOP 3.87 average climate, flow temperature 60°C maximum, 3-phase 400V, 38A max current, ODU 1,680×1,240×680 mm, ODU weight 290 kg, sound power 72 dB(A), MCS certified, PAC-IF013B hydraulic module

STRENGTHS

Very strong UK commercial installer network; PAC hydraulic module well-specified with buffer vessel and controls; excellent UK technical support; 5-year warranty standard; proven reliability data

WEAKNESSES

R410A – highest GWP in class, F-Gas phase-down risk; 45 kW minimum over-sizes for 30 kW buildings; very high weight (290 kg ODU) requires crane for rooftop; price significantly above target market; no sub-35 kW commercial product in UK range

DIFFERENTIATION ANGLE

Our 30 kW product fills Mitsubishi's gap in the 25–35 kW commercial segment at less than half the price, with R290 eliminating F-Gas regulatory exposure for building owners

Vaillant Group

Product	aroTHERM plus VWL 35/5 AS (5.7–9.3 kW) – residential; no commercial 30 kW R290 product
Pricing	£2,100–£2,600 supply-only (trade, UK 2023) for ODU; uni-TOWER hydraulic unit additional £800–£1,000
Tech Specs	R290 refrigerant, 9.3 kW at A2/W35, COP 4.2 at A2/W35 (EN 14825), SCOP 5.0 average climate at W35, flow temperature 75°C maximum, single-phase 230V, ODU 1,168×488×400 mm, ODU weight 81 kg, 40 dB(A) at 1m, MCS certified, companion uniTOWER hydraulic station

STRENGTHS

Only major brand with MCS-certified R290 product in UK market; proven R290 design with ATEX-compliant enclosure; excellent SCOP performance; strong brand for residential; useful benchmark for our R290 engineering approach

WEAKNESSES

Maximum 9.3 kW – cannot address commercial 30 kW demand even in multiple units; residential product architecture not suited to commercial hydronic integration; 230V single-phase limits commercial application; no commercial support structure or commercial warranty terms

DIFFERENTIATION ANGLE

We scale Vaillant's proven R290 approach to the commercial 30 kW segment with 3-phase architecture, commercial hydronic integration, Modbus controls, and MCS commercial product certification – a market Vaillant does not serve

Viessmann

Product	Vitocal 300-A AWO-E-AC 301.B16 (16 kW at A2/W35, up to 35 kW cascade)
Pricing	£5,200–£6,100 single ODU supply-only (UK trade 2023); cascade pair £9,800–£11,500
Tech Specs	R290 refrigerant, 16 kW rated at A2/W35, COP 3.9 at A2/W35, flow temperature 70°C max, 3-phase 400V, ODU 1,330×1,100×415 mm, weight 176 kg, 55 dB(A), MCS certified UK, companion Hydronic Module WO1 available, cascade to 35 kW with two ODUs

STRENGTHS

R290 with MCS certification already achieved – first mover in commercial R290 in UK; 70°C flow temperature for HT retrofit; strong German engineering reputation; cascade option reaches 32–35 kW; good European commercial service network

WEAKNESSES

Single unit only 16 kW – requires cascade for 30 kW, doubling refrigerant circuits and ODU footprint; cascade hydraulic complexity; premium pricing; ODU at 176 kg is heavy for wall-mount; limited UK commercial installer training program vs Daikin/Mitsubishi

DIFFERENTIATION ANGLE

Our single-unit 30 kW R290 solution eliminates the cascade complexity of Viessmann's approach, reducing installation cost by an estimated £800–£1,200 in labour and pipework while achieving equivalent or superior COP in a simpler architecture at lower supply cost

NIBE Energy Systems

Product	NIBE F2120-20 (20 kW) Air Source Heat Pump
Pricing	£4,400–£5,200 supply-only ODU (UK trade 2023); SMO controller and indoor module additional £1,100–£1,500
Tech Specs	R410A refrigerant (GWP 2,088), 20 kW at A2/W35, COP 3.59 at A2/W35, SCOP 3.82 average climate, flow temperature 65°C, 3-phase 400V, ODU 1,620×630×640 mm, weight 152 kg, 66 dB(A) sound power, MCS certified, NIBE SMO S40 controller with BMS integration, compatible with NIBE VVM S320 indoor module

STRENGTHS

Closest competitor to our target price point; strong Scandinavian cold-climate engineering pedigree; good BMS/smart-home integration; MCS certified; reasonable UK installer network; 20 kW available as single unit

WEAKNESSES

R410A – F-Gas phase-down risk; maximum 20 kW from single unit (our 30 kW target requires 1.5× capacity); high sound power level 66 dB(A) may preclude noise-sensitive planning sites; no R290 transition roadmap publicly announced for commercial range; software ecosystem relatively closed

DIFFERENTIATION ANGLE

Our 30 kW R290 unit matches NIBE's price bracket while delivering 50% more capacity from a single unit with zero F-Gas exposure, open Modbus protocol, and a lower GWP that future-proofs the building owner's investment

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£4,500.00
Max Mass	180 kg

Research Sources

Title	Type	Year	Relevance
BEIS Heat Pump Market Report 2023	government_report	2026	UK market volume data, 8,000 non-domestic units/year baseline figure cited
IEA The Future of Heat Pumps 2022	industry_report	2026	Global commercial heat pump CAGR 18%, technology roadmap data
BS EN 378-1:2016+A1:2020 Refrigerating systems and heat pumps – Safety and environmental requirements Part 1	standard	2026	R290 charge limits (2.5 kg occupied space), safety classification basis
BS EN 14825:2022 Air conditioners, liquid chilling packages and heat pumps – Testing and rating at part load conditions	standard	2026	SCOP calculation methodology and test point definition including A2/W35
MCS MCS 007 Issue 3.0 – Heat Pump Products Certification Scheme	certification_standard	2026	Mandatory product certification route for UK BUS eligibility
MCS MIS 3005 Issue 6.0 – Installer Standard for Heat Pump Systems	certification_standard	2026	Installer requirements and system design standards for MCS-certified installations
ASHRAE 34-2022 Designation and Safety Classification of Refrigerants	standard	2026	R290 Class A3 flammability classification, LFL/UFL data
DESNZ UK Government Greenhouse Gas Conversion Factors 2023	government_report	2026	Grid carbon intensity 0.207 kg-CO ₂ e/kWh used in carbon saving calculations

Copeland Scroll Compressor Engineering Guide R290 Series 2022	manufacter_datasheet	2026	ZH26KVE compressor displacement, R290 compatibility, performance maps at A2/W35
SWEP BPHE Selection Handbook 2023	manufacter_datasheet	2026	B25T series plate count, UA values, pressure rating for condenser sizing
Danfoss Refrigeration & Air Conditioning ETS EXV Product Manual 2023	manufacter_datasheet	2026	ETS 12.5 valve capacity and control parameters for R290 at specified operating conditions
HSE DSEAR 2002 – Dangerous Substances and Explosive Atmospheres Regulations	regulation	2026	ATEX zone classification obligations for R290 charge in commercial premises
PED 2014/68/EU (retained in UK as SI 2016/1105) Pressure Equipment Directive	regulation	2026	Pressure equipment category classification for refrigerant circuit and hydraulic components

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2.1 BS EN 378-1:2016+A1:2020 [D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements – Part 1: Basic requirements, definitions, classification and selection criteria

Status: not-started

Owner: Lead Mechanical/Refrigeration Engineer

Summary

Defines safety requirements for refrigerating systems, refrigerant classifications, charge limits in occupied spaces, and risk assessment methodology

Applicability

Governs R290 charge quantity (d2.5 kg in occupied spaces), system boundary definitions between ODU/IHU, refrigerant containment, and emergency shut-off requirements for the complete split system

Engineering Impact

Forces all refrigerant-containing components to reside in the outdoor unit or a dedicated plant room; IHU must be refrigerant-free (water-side only); requires automatic leak detection system if refrigerant crosses indoor boundary; ODU enclosure must be ventilated per Annex C calculations for A3 refrigerant

EVIDENCE REQUIRED

Third-party assessed risk assessment per Clause 4; material compatibility declarations; pressure test records at $1.1 \times PS$ (minimum 33 bar for R290 circuit); charge quantity calculation documented in technical file

GAP ACTION

Commission refrigeration risk assessment (EN 378-1 Annex B) and ODU enclosure ventilation calculation; engage notified body for Cat II PED file review simultaneously

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.2 BS EN 378-2:2016+A1:2020 ^[D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements – Part 2: Design, construction, testing, marking and documentation

Status: not-started

Owner: Lead Mechanical/Refrigeration Engineer

Summary

Prescribes construction standards, pressure equipment design, relief valve sizing, piping, joint types, and documentation requirements for the refrigeration circuit

Applicability

Applies to all brazed and flared joints in R290 circuit, relief valve on high side, sight glass specification, service valve requirements, and technical documentation package for UKCA marking

Engineering Impact

Brazed joints must use silver alloy brazing (minimum 30% Ag) per Clause 6; all pipe material to be dehydrated seamless copper to EN 12735-1; high-side relief valve set at dPS (25 bar design pressure); suction accumulator required if compressor flood risk exists at low ambient

EVIDENCE REQUIRED

Pressure test certificate at $1.1 \times$ design pressure; vacuum dehydration log (target <270 Pa residual); weld/braze procedure qualification records; relief valve sizing calculation per Annex D

GAP ACTION

Prepare refrigerant circuit P&ID to EN 378-2 Annex E format; procure R290-rated relief valve (Castel 3060 series or equivalent) and size per standard annex

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.3 BS EN 378-3:2016+A1:2020 [D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements – Part 3: Installation site and personal protection

Status: not-started

Owner: Applications/Installation Engineering Manager

Summary

Covers installation requirements, ventilation of machinery rooms, personal protective equipment, and emergency procedures for installers

Applicability

Defines ventilation requirements for any enclosed plant room housing the ODU or containing refrigerant, and minimum separation distances; relevant to installation manual content and MCS MIS 3005 compliance

Engineering Impact

Installation manual must specify minimum clearances, plant room ventilation rate formula (EN 378-3 Eq. 1 based on charge and LFL), and requirement for fixed gas detection where ODU is in semi-enclosed space; forces inclusion of Schraeder service valves on ODU for safe charging

EVIDENCE REQUIRED

Installation manual reviewed against Part 3 requirements; sample installation sign-off checklist; gas detector specification and placement diagram in technical documentation

GAP ACTION

Draft installation manual chapter on R290 safety, plant room ventilation calculation template, and gas detector product recommendation list

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.4 BS EN 14825:2022 ^[D]

Air Conditioners, Liquid Chilling Packages and Heat Pumps – Testing and Rating at Part Load Conditions Including Low Temperature Testing

Status: not-started

Owner: Test & Certification Manager

Summary

Specifies test conditions, test procedures, and SCOP/SEER calculation methodology for heat pumps, including the bin-hours method for UK/European climates

Applicability

Mandatory for determining SCOP value reported on ErP product data sheet and for MCS MCS 007 product submission; A2/W35 rating point is one of seven required test points for SCOP calculation (EN 14825 Table 3, average climate)

Engineering Impact

Forces system to operate and be tested at -7°C, 2°C, 7°C, 12°C outdoor air conditions; degradation coefficient Cd must be measured or defaulted at 0.25; cycling losses must be characterised; controls must support stable operation across full test matrix

EVIDENCE REQUIRED

Full EN 14825 test report from UKAS-accredited laboratory covering all 7 rating points; Cd measurement; SCOP calculation worksheet; ErP product fiche completed to EU 813/2013 Regulation format (retained UK equivalent)

GAP ACTION

Identify UKAS-accredited test house (e.g. BSRIA, TÜV SÜD, Intertek Letchworth) with R290 calorimeter capability; book pre-compliance test slot 6 months prior to certification submission

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.5 MCS MCS 007 Issue 3.0 [D]

MCS Heat Pump Products Certification Scheme Requirements

Status: not-started

Owner: Certification & Quality Manager

Summary

MCS product certification scheme mandatory for heat pumps installed under BUS grants and for installer MCS compliance; requires independent testing and annual audits

Applicability

Product must be listed on MCS Product Directory before any BUS-qualifying installation; requires minimum SCOP data at three UK climate zones (average, warmer, colder per EN 14825) and full technical documentation submission

Engineering Impact

Forces completion of EN 14825 testing before market launch; requires documented quality management system for production; mandates recall and corrective action procedure; serial number and UKCA marking on each unit; digital product passport emerging requirement from 2025

EVIDENCE REQUIRED

MCS 007 application form; EN 14825 test report; ISO 9001 or equivalent QMS certificate for manufacturing site; technical file including risk assessment, wiring diagrams, and installation manual; product insurance/liability documentation

GAP ACTION

Submit MCS 007 pre-application to MCS Scheme Administrator; appoint MCS-approved certification body (e.g. Kiwa, BRE, CERTASUN); initiate ISO 9001 gap analysis for manufacturing process

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	thermal_system

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
r290_vapor_compression_loop	2236 × 2236 × 2400	5	floor
odu_ambient_heat_absorption_module	2236 × 2236 × 2400	5	floor
indoor_heat_rejection_bphe	2236 × 2236 × 2400	5	floor
precision_refrigerant_expansion_system	2236 × 2236 × 2400	5	floor
primary_hydraulic_circulation_loop	2236 × 2236 × 2400	5	floor
power_electronics_inverter_drive	2236 × 2236 × 2400	5	floor
main_control_bms_interfacing_unit	2236 × 2236 × 2400	5	floor
odu_enclosure_atex_safety_system	2236 × 2236 × 2400	5	floor
reverse_cycle_defrost_management	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (45.0m²) exceeds envelope budget (21.3m²)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 R290 Vapor Compression Loop [D]

Maturity: draft

Purpose

Compresses approximately 0.068 kg/s of R290 vapor from an evaporating pressure corresponding to -8°C up to a condensing pressure at 42°C, providing a 50 K thermodynamic lift.

Why It Matters

Not computed

Technical Description

The sealed outdoor refrigerant circuit conforms to BS EN 378 and contains 2.0 kg of R290 (propane, A3 flammability class) , remaining below the 2.5 kg occupied-space charge limit. The scroll compressor operates at 2,900 rpm with a target swept volume of 32–38 cm³/rev . Silver-alloy brazing with a minimum 30% Ag composition joins all copper pipework to EN 378-2 requirements, providing leak-resistant joints at all 8–12 brazed connections . The circuit design pressure is PS 25 bar, requiring PED 2014/68/EU Category II classification due to R290's Group 1 hazard designation. All pressure-containing components undergo factory proof-testing at 33 bar (1.32 × PS) . At low ambient conditions where COP reduces to approximately 2.5 , a suction accumulator upstream of the compressor retains any unevaporated liquid, preventing liquid carryover exceeding 5% vapour quality at suction. A liquid receiver downstream of the condenser maintains sub-cooled liquid feed to the expansion valve under all transient load conditions. Direct material cost for this assembly is £900–£1,050 per unit at a production batch of 500 .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 ODU Ambient Heat Absorption Module [D]

Maturity: draft

Purpose

Extracts low-grade latent and sensible heat from outdoor ambient air using cross-flow forced convection, evaporating liquid R290 at -8°C target.

Why It Matters

Not computed

Technical Description

The outdoor evaporator is a louvred-fin aluminium microchannel coil with a face area of 0.9 m² and a fin pitch of 1.4 mm . The microchannel construction limits internal refrigerant volume to between 1.8 kg and 2.2 kg total system charge , remaining below ADR dangerous-goods road-transit thresholds that apply above 333 kg for Group 2 LPG. Two 560 mm EC axial fans deliver 3,500 m³/h combined airflow against 40 Pa external static pressure . Fan power consumption ranges from 180 W at minimum inverter speed to 450 W at full speed . The coil assembly costs £280–£340 and the fan pair costs £220–£280 at 500-unit batch quantities . A BS EN ISO 13857-compliant fan guard prevents finger ingress, with mesh apertures not exceeding 12 mm across any opening, satisfying UK Machinery Directive SI 2016/1091.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 Indoor Heat Rejection BPHE [D]

Maturity: draft

Purpose

Transfers thermal energy from 42°C condensing R290 vapour into the primary water loop, delivering 30 kW of heating capacity.

Why It Matters

Not computed

Technical Description

The condenser is a 40-plate brazed plate heat exchanger (BPHE) constructed from 316L stainless steel plates with copper brazing, achieving a design UA_cond of 4.2 kW/K . The unit is co-located within the ODU rear partition, ensuring the indoor hydraulic unit contains no refrigerant-side components and maintains strict EN 378 zone separation. Primary water circulates through the secondary channels at 1.3 l/s , entering at 29.5°C and leaving at 35.0°C, a temperature rise of 5.5 K . The refrigerant-side design pressure rating is PS 30 bar, verified by hydraulic pressure test at 45 bar . R290 leaves the condenser sub-cooled by 3–5 K to ensure single-phase liquid feed to the expansion valve and prevent vapour lock. The BPHE is insulated with a 19 mm closed-cell elastomeric foam jacket, reducing parasitic heat loss to ambient by an estimated 8–12 W at a 10 K temperature differential . Ex-works unit cost is £180–£220 at 500-unit batch volume .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Precision Refrigerant Expansion System [D]

Maturity: draft

Purpose

Actively meters liquid R290 into the evaporator, expanding it from 15 bar condensing pressure down to 3 bar evaporating pressure while maintaining 5–8 K superheat.

Why It Matters

Not computed

Technical Description

The electronic expansion valve (EXV) reduces R290 pressure from 15 bar to 3 bar across a modulating orifice driven by a unipolar stepper motor with a resolution of approximately 480 steps full-stroke . The control system polls a suction-line ratiometric pressure transducer and an NTC thermistor at 0.5-second intervals, computing suction superheat to ± 0.5 K resolution . A maximum operating pressure (MOP) function limits suction pressure to 5.5 bar during cold-start conditions , keeping compressor motor current below the 16 A protection threshold. The EXV targets 5–8 K superheat at the evaporator outlet ; superheat above 10 K causes the final 15–20% of the 0.9 m² evaporator face area to perform sensible-only heat transfer, reducing capacity . Superheat below 3 K risks unevaporated R290 droplets reaching compressor suction. Total subsystem cost, including valve, stepper driver, and sensors, is £150–£190 at 500-unit batch volume .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 Primary Hydronic Circulation Loop [D]

Maturity: draft

Purpose

Decouples and circulates primary system water at 1.3 l/s, moving thermal energy from the ODU into a 50–80 litre buffer vessel inside the building.

Why It Matters

Not computed

Technical Description

The indoor hydraulic unit (IHU) houses the primary circulation loop, which operates at flow temperatures up to 65°C , with the EN 14825 A2/W35 design condition specifying 35°C flow and 30°C return. A variable-speed IE7-class circulator pump operates from a 230 V single-phase supply, consuming between 5 W and 45 W across its duty range . Internal pipework uses 28 mm copper or 316 stainless steel, connected through a hydraulic low-loss header to isolate the primary 1.3 l/s fixed-speed circuit from variable-flow secondary zone circuits. The 50–80 L buffer vessel provides sufficient thermal mass to enforce a minimum compressor run time of 10 minutes , preventing short-cycling. During a 4-minute reverse-cycle defrost event, the buffer releases 3–5 kWh of stored heat to maintain building radiator temperatures. Assembly cost is £280–£350 per unit at 500-unit batch volume .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Power Electronics & Inverter Drive [D]

Maturity: draft

Purpose

Converts 3-phase 400 V grid AC into a variable waveform to modulate compressor capacity, smoothing inrush current and limiting total active power to approximately 8.57 kW.

Why It Matters

Not computed

Technical Description

The inverter drive accepts a 3-phase 400 V, 50 Hz supply protected by a 16 A type-C MCB and generates a variable-frequency synthetic sine wave using insulated-gate bipolar transistors (IGBTs) switching at 4–16 kHz . Compressor speed varies between 1,200 rpm and 4,500 rpm under inverter control . The module complies with BS EN 61000-6-2 for industrial immunity and BS EN 61000-6-4 for conducted and radiated emissions. A built-in Class C2 EMC filter comprising common-mode chokes and X/Y capacitors attenuates high-frequency noise above 150 kHz from the building supply . The inverter housing is mounted in a dedicated steel-bulkheaded compartment, physically separated from the R290 refrigerant zone by a minimum 1.5 mm steel partition , preventing spark exposure to a potentially explosive atmosphere. Unit cost is £600–£750 at 500-unit batch volume .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 Main Control & BMS Interfacing Unit [D]

Maturity: draft

Purpose

Executes thermodynamic PID loops, safety fault management, EN 14825 SCOP degradation mitigation, and interfaces with the building's energy management system over Modbus/BACnet.

Why It Matters

Not computed

Technical Description

The control board, based on a Carel pCO5 platform or an equivalent STM32-based design, manages a minimum of 32 I/O points across the ODU and IHU , sampling pressure transducers and thermistors at intervals of 100 ms or faster . SCOP performance under EN 14825 part-load conditions depends on the board's capacity to modulate compressor speed smoothly at 50% load on a 7°C ambient day , minimising the degradation coefficient Cd. External weather compensation set-point data is transmitted over a shielded RS485 bus via Modbus RTU to building energy management systems covering floor areas of 300–600 m² . To comply with IEC 60335-2-40:2022 safety requirements for A3 refrigerants, the controller enforces a 30-second fan pre-purge sequence before closing any high-voltage contactor, dispersing any pooled R290 vapour at ground level. The 24 V DC logic supply is derived from an onboard SMPS rated at 60 W .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 ODU Enclosure & ATEX Safety System [D]

Maturity: draft

Purpose

Provides physical weather protection, acoustic attenuation to 57–66 dBA, and R290 leak containment with active ventilation conforming to HSE DSEAR 2002.

Why It Matters

Not computed

Technical Description

The outdoor unit chassis is fabricated from 1.5 mm powder-coated galvanised steel sheet , enclosing an assembly mass exceeding 150 kg . Sheet metal and acoustic baffling material costs £450–£550 per unit at 500-unit batch volume . The compressor compartment base incorporates louvred ventilation apertures to drain propane vapour, which has a density of approximately 2.0 kg/m³ at 15°C , significantly greater than air at 1.2 kg/m³ , causing leaks to pool at the lowest point. An MSA Bacharach fixed infrared gas detector monitors propane concentration continuously at base level. When the detector reads above 10% of the lower explosive limit (LEL, equivalent to approximately 2,100 ppm by volume for propane) , solid-state relays cut power to the compressor within 500 ms , a visual strobe activates, and EC fans ramp to 100% duty to disperse vapour. This response sequence maintains compliance with DSEAR 2002 and the ATEX zoning requirements applicable to the immediate compressor enclosure.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Reverse-Cycle Defrost Management [D]

Maturity: draft

Purpose

Detects frost accumulation on the 1.4 mm fin pitch evaporator and reverses refrigerant flow to inject 42°C discharge gas, melting accumulated ice in under 5 minutes.

Why It Matters

Not computed

Technical Description

Frost accumulates on the 0.9 m² evaporator face at a rate sufficient to require defrost intervention when ambient conditions fall between -2°C and +4°C at relative humidity above 80% . The control board initiates defrost when suction pressure drops more than 1.5 bar below the ambient-corrected saturation value, or when the coil surface temperature diverges more than 4 K below ambient . Target defrost frequency is fewer than 4 events per 24-hour period . The Sanhua 4-way reversing valve shifts state, routing high-pressure R290 vapour at 42°C directly into the outdoor coil, which temporarily acts as a condenser. Simultaneously, the indoor BPHE draws 4–6 kW of heat from the 50–80 L hydronic buffer , acting as a temporary evaporator. To reduce hydraulic shock during valve transition, the compressor speed reduces to 1,200 rpm for 10 seconds before the reversing valve shifts , limiting the pressure transient to below 3 bar/s . Total defrost cycle duration is 3–5 minutes .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
R290 Vapor Compression Loop	R290 catastrophic leak from compressor housing rupture	Not computed	Not computed	5×1=5	PED Cat II qualification and 33 bar hydrostatic pressure testing during assembly.
R290 Vapor Compression Loop	Refrigerant fluid migration during off-cycle	Not computed	Not computed	4×3=12	Install a 40W silicone crankcase heater interlocked with ambient thermistor.
R290 Vapor Compression Loop	Overpressurization on high-side	Not computed	Not computed	5×2=10	Inline Castel pressure relief valve set at exactly 25 bar relieving to secure outdoor zone.
ODU Ambient Heat Absorption Module	Fan blade contact causing injury	Not computed	Not computed	4×1=4	Mesh fan guard sized specifically to EN ISO 13857 minimum gap sizing.
ODU Ambient Heat Absorption Module	Inadequate airflow reducing target 30kW limit	Not computed	Not computed	3×3=9	Variable speed EC fan control logic tied to suction pressure transducer.
ODU Ambient Heat Absorption Module	Corrosive fin rot in UK coastal bands	Not computed	Not computed	3×2=6	Specify factory-applied e-coating or hydrophilic epoxy coating over entire coil.
Indoor Heat Rejection BPHE	Water side freezing and bursting BPHE	Not computed	Not computed	5×2=10	Antifreeze thermistor on water leaving port; soft-lockout if T < 4°C.
Indoor Heat Rejection BPHE	Fouling causing capacity drop	Not computed	Not computed	3×4=12	Mandate magnetic dirt separator on IHU water inlet in installer manual.

Indoor Heat Rejection BPHE	Heat loss to ambient air	Not computed	Not computed	2×5=10	Install 19mm elastomeric closed-cell foam tightly sealed with V-tape.
Precision Refrigerant Expansion System	Liquid flood back to compressor	Not computed	Not computed	5×2=10	Implement fast-close logic on the Carel driver if superheat drops below 2 K for 5 seconds.
Precision Refrigerant Expansion System	Internal moisture freezing ring	Not computed	Not computed	4×2=8	Enforce strict vacuum dehydration log (target <270 Pa) during factory assembly.
Precision Refrigerant Expansion System	Electrical connector corrosion on EXV head	Not computed	Not computed	3×3=9	Use IP67 rated molded DIN connectors with dielectric grease applied at factory.
Primary Hydronic Circulation Loop	Loss of water flow during operation	Not computed	Not computed	4×3=12	Hardwired SIKA flow switch directly cuts compressor contactor circuit independently of software.
Primary Hydronic Circulation Loop	System freezing during winter power-outage	Not computed	Not computed	5×1=5	Recommend 20% propylene glycol mix in installer manuals for exposed pipe runs.
Primary Hydronic Circulation Loop	Insufficient water volume causing short-cycling	Not computed	Not computed	3×4=12	Include internal 50L buffer tank standard to force minimum thermal mass.
Power Electronics & Inverter Drive	Spark igniting an R290 leak	Not computed	Not computed	5×1=5	Design physical steel bulkhead between power electronics compartment and compressor bay.
Power Electronics & Inverter Drive	Conducted EMI failing CE/UKCA EMC testing	Not computed	Not computed	4×3=12	Specify built-in Class C2 filter and mandate shielded motor cables earthed both ends.
Power Electronics & Inverter Drive	Inverter overheat trip at 35°C ambient	Not computed	Not computed	3×2=6	Expose inverter heatsink out of the sealed PCB housing directly into the EC fan draft stream.

Main Control & BMS Interfacing Unit	A3 ignition sequence violation	Not computed	Not computed	5×2=10	Hard-code the pre-purge fan delay using physical, non-programmable time-delay re-lays as backup.
Main Control & BMS Interfacing Unit	Lightning surge over communication cable	Not computed	Not computed	3×2=6	Install optical isolation on the RS485 port and TVS diodes on all sensor inputs.
Main Control & BMS Interfacing Unit	Sensor failure causing extreme water temps	Not computed	Not computed	4×2=8	Dual redundancy on flow temperature sensors; compare values and trip if deviation >2 K.
ODU Enclosure & ATEX Safety System	Propane pooling in compressor bay base	Not computed	Not computed	5×2=10	Design louvered baseplate sloped at 2 degrees to naturally drain heavy gas in zero-wind conditions.
ODU Enclosure & ATEX Safety System	Acoustic resonance breaking 57 dBA limit	Not computed	Not computed	2×4=8	Apply heavy bitumen deadening pads to all major 1.5mm flat steel panels.
ODU Enclosure & ATEX Safety System	Gas detector failure during active leak	Not computed	Not computed	5×1=5	Specify fail-safe circuitry; loss of heartbeat from detector defaults system into continuous purge mode.
Reverse-Cycle Defrost Management	Water-side freezing in BPHE during defrost	Not computed	Not computed	5×2=10	Immediate termination of defrost sequence if BPHE leaving water drops below 15°C.
Reverse-Cycle Defrost Management	Compressor liquid slugging on valve swap	Not computed	Not computed	4×3=12	Suction accumulator buffers the sudden liquid return; EXV forcefully throttles to 10% momentarily.
Reverse-Cycle Defrost Management	Ice damming at base of ODU pan	Not computed	Not computed	3×4=12	Route hot liquid discharge line directly through the base drainage pan before the EXV.

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	9 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success

Section Evaluation: Brief

9/100

Use case is detailed	50/100	Use case is detailed
Mission statement present	50/100	Mission statement present
Target customers defined	50/100	Target customers defined
Market timing rationale present	50/100	Market timing rationale present
Cost ceiling declared	50/100	Cost ceiling declared
Mass constraint declared	50/100	Mass constraint declared

Recommended Code Changes:

Add compliance notes

Section Evaluation: Regulatory

8/100

10 standards identified	50/100	10 standards identified
Comprehensive regulatory coverage	50/100	Comprehensive regulatory coverage
Most standards have applicability detail	50/100	Most standards have applicability detail
Most standards have design impact analysis	50/100	Most standards have design impact analysis
Gap actions defined for some standards	50/100	Gap actions defined for some standards

Section Evaluation: Sizing

6/100

Design is infeasible — shown as exception	50/100	Design is infeasible — shown as exception
Module dimensions computed	50/100	Module dimensions computed
1 conflicts identified	50/100	1 conflicts identified
Recommendations provided	50/100	Recommendations provided

Section Evaluation: Modules

8/100

9 modules — good granularity	50/100	9 modules — good granularity
Most modules have detailed descriptions	50/100	Most modules have detailed descriptions
Most modules have 3+ key parts	50/100	Most modules have 3+ key parts
Most modules have risk matrices	50/100	Most modules have risk matrices

Recommended Code Changes:
More modules need engineering reviews

Section Evaluation: BOM

1/100

No parts generated 50/100 No parts generated

Recommended Code Changes:

Run BOM generation

Section Evaluation: Cost

1/100

No cost breakdown 50/100 No cost breakdown

Recommended Code Changes:

Run cost calculation

Section Evaluation: Risks

7/100

27 risks identified — comprehensive	50/100	27 risks identified — comprehensive
Most risks have mitigations	50/100	Most risks have mitigations
14 high-priority risks (score >= 10)	50/100	14 high-priority risks (score >= 10)

Recommended Code Changes:

Assign owners to more risks

Section Evaluation: Suppliers

2/100

No supplier matches50/100No supplier matches

Recommended Code Changes:

Set BRAVE_API_KEY and re-run supplier stage

Section Evaluation: Research

7/100

Research report: 5444 chars	50/100	Research report: 5444 chars
13 sources cited	50/100	13 sources cited
Domain detected: thermal_system	50/100	Domain detected: thermal_system